

B. Fun Permutation

time limit per test: 2 seconds

memory limit per test: 256 megabytes

You are given a permutation* p of size n .

Your task is to find a permutation q of size n such that $\text{GCD}^{\dagger}(p_i + q_i, p_{i+1} + q_{i+1}) \geq 3$ for all $1 \leq i < n$. In other words, the greatest common divisor of the sum of any two adjacent positions should be at least 3.

It can be shown that this is always possible.

*A permutation of length m is an array consisting of m distinct integers from 1 to m in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($m = 3$ but there is 4 in the array).

$^{\dagger}\text{gcd}(x, y)$ denotes the [greatest common divisor \(GCD\)](#) of integers x and y .

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($2 \leq n \leq 2 \cdot 10^5$).

The second line contains n integers $p_1, p_2, \dots, p_n$ ($1 \leq p_i \leq n$).

It is guaranteed that the given array forms a permutation.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the permutation q on a new line. If there are multiple possible answers, you may output any.

Example

input	Copy
<pre>3 3 1 3 2 5 5 1 2 4 3 7 6 7 1 5 4 3 2</pre>	
output	Copy
<pre>2 3 1 4 5 1 2 3 2 1 3 7 5 6 4</pre>	

Note

In the first test case, $\text{GCD}(1 + 2, 3 + 3) = 3 \geq 3$ and $\text{GCD}(3 + 3, 2 + 1) = 3 \geq 3$, so the output is correct.

Codeforces Round 1047 (Div. 3)

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Submit?

Language:

Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file:

Choose File

No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
337427696	Sep/07/2025 19:28	Accepted
337422014	Sep/07/2025 19:22	Wrong answer on test 1
337403286	Sep/07/2025 19:01	Wrong answer on test 1
337401464	Sep/07/2025 18:59	Runtime error on test 1

→ Problem tags

constructive algorithms

math

number theory

No tag edit access

→ Contest materials

- Announcement (en)